

QUANTITATIVE METHODS I · WEEK 0

Levelling up

Everything the course assumes you already have: powers, expanding, factoring, fractions, equations, inequalities and systems.

- 0.1 Powers and roots
 - 0.2 Working with algebraic expressions
 - 0.3 Factoring
 - 0.4 Fractions
 - 0.5 Linear equations
 - 0.6 Quadratic equations
 - 0.7 Turning a situation into an equation
 - 0.8 Inequalities and percentages
 - 0.9 Systems of linear equations
 - 0.10 Test yourself
-

0.1 Powers and roots

- 1 Same base, multiplied: add the exponents.

$$a^m \cdot a^n = a^{m+n}$$

- 2 Same base, divided: subtract them.

$$\frac{a^m}{a^n} = a^{m-n}$$

- 3 A power of a power: multiply them.

$$(a^m)^n = a^{mn}$$

- 4 A product inside a power: the power reaches every factor.

$$(ab)^n = a^n b^n$$

- 5 Zero and negative exponents are definitions, not tricks.

$$a^0 = 1 \quad (a \neq 0) \quad a^{-n} = \frac{1}{a^n}$$

6 A root is a fractional exponent, which is why the same five rules cover roots.

$$\sqrt[n]{a} = a^{1/n} \quad \text{the } n\text{-th root of } a = a^{1/n}$$

Simplify $\frac{(2x^3)^4}{4x^5}$

$$\frac{16x^{12}}{4x^5}$$

The fourth power reaches the 2 as well as the x^3 , so $2^4 = 16$ and x^{12} .

$$4x^7$$

Divide the numbers, subtract the exponents.

WATCH OUT

- $(a + b)^2$ is not $a^2 + b^2$. A power does not reach across a plus.
- a^{-n} is not negative. It is a reciprocal, so $2^{-3} = \frac{1}{8}$.
- $\sqrt{a^2 + b^2}$ does not split into $a + b$. Test it at $a = b = 3$.

LEARN IT ON KHAN ACADEMY

Video The five rules · Exponent properties review

Algebra 1, unit 11: Exponents and radicals

Video Simplifying square roots

Algebra 1, unit 11: Exponents and radicals

PRACTISE IT ON KHAN ACADEMY

Exercise The rules together · Properties of exponents challenge (integer exponents)

Algebra 1, unit 11: Exponents and radicals

Exercise Simplify square roots

Algebra 1, unit 11: Exponents and radicals

0.2 Working with algebraic expressions

Term. What the + and – signs separate. In $3x^2 - 5x + 2$ there are three terms.

Factor. What multiplies the whole of something. Factors cancel; terms never do.

Like terms. Same letter, same power. $3x$ and $5x$ are alike; $3x$ and $3x^2$ are not.

1 Distribute across every term in the bracket.

$$a(b + c) = ab + ac$$

- 2 **Two brackets: every term of the first against every term of the second.**

$$(a + b)(c + d) = ac + ad + bc + bd$$

- 3 **Two patterns are worth knowing by sight.**

$$(a + b)(a - b) = a^2 - b^2 \quad (a \pm b)^2 = a^2 \pm 2ab + b^2$$

- 4 **A minus in front of a bracket reaches every term inside it.**

$$-(x - 4) = -x + 4$$

- 5 **Collect like terms last.** Nothing else combines.

Expand $(2x - 5)(3x + 4)$

$$6x^2 + 8x - 15x - 20$$

Each term of the first bracket against each term of the second.

$$6x^2 - 7x - 20$$

Collect the two x terms.

WATCH OUT

- $2(x + 3)$ is $2x + 6$, not $2x + 3$. The 2 reaches both terms.
- $(q + 3)^2$ is $q^2 + 6q + 9$. The middle term is the one people drop.
- $3x$ and $3x^2$ do not combine. Different power, different term.

LEARN IT ON KHAN ACADEMY

Video Expanding, from the start · Multiplying binomials intro

Algebra 1, unit 13: Quadratics, multiplying and factoring

Video Two brackets · Multiplying binomials

Algebra 1, unit 13: Quadratics, multiplying and factoring

Article Worked through · Warmup: Multiplying binomials

Algebra 1, unit 13: Quadratics, multiplying and factoring

PRACTISE IT ON KHAN ACADEMY

Exercise Expanding · Multiply binomials

Algebra 1, unit 13: Quadratics, multiplying and factoring

Exercise The difference of two squares · Multiply difference of squares

Algebra 1, unit 13: Quadratics, multiplying and factoring

0.3 Factoring

- 1 Take out the common factor first, every time.

$$6x^2 + 9x = 3x(2x + 3)$$

- 2 For $x^2 + bx + c$, find two numbers that multiply to c and add to b .

$$x^2 - 13x + 36 = (x - 4)(x - 9)$$

- 3 Difference of two squares.

$$a^2 - b^2 = (a - b)(a + b)$$

- 4 Perfect square.

$$a^2 \pm 2ab + b^2 = (a \pm b)^2$$

- 5 When the leading coefficient is not 1, split the middle term and group. Find two numbers multiplying to ac and adding to b .
- 6 Check by expanding. It costs ten seconds and catches every sign error.

Factor $3x^2 + 11x - 4$

$$ac = 3 \times (-4) = -12$$

Two numbers multiplying to -12 and adding to 11 : 12 and -1 .

$$3x^2 + 12x - x - 4$$

Split the middle term using those two numbers.

$$3x(x + 4) - 1(x + 4)$$

Group in pairs and take each common factor out.

$$(3x - 1)(x + 4)$$

The bracket $x + 4$ is now itself the common factor.

WATCH OUT

- $x^2 + 9$ does not factor over the real numbers. A sum of two squares never does.
- Missing the common factor first makes every later step harder and sometimes impossible.
- $4x^2 - 16$ is $4(x - 2)(x + 2)$. Completely means completely.

LEARN IT ON KHAN ACADEMY

Video What factoring is · Introduction to factoring

Algebra 1, unit 13: Quadratics, multiplying and factoring

Video Factoring a quadratic · Factoring quadratics intro

Algebra 1, unit 13: Quadratics, multiplying and factoring

Video The difference of two squares · Factoring quadratics with difference of squares

Algebra 1, unit 13: Quadratics, multiplying and factoring

Article The method, written out · Factoring simple quadratics review

Algebra 1, unit 13: Quadratics, multiplying and factoring

PRACTISE IT ON KHAN ACADEMY

Exercise Common factor first · GCF factoring introduction

Algebra 1, unit 13: Quadratics, multiplying and factoring

Exercise Factoring a quadratic · Factoring quadratics intro

Algebra 1, unit 13: Quadratics, multiplying and factoring

0.4 Fractions

- 1 **Multiply straight across. No preparation needed.**

$$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$$

- 2 **Divide by turning the second one over.**

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}$$

- 3 **Only adding and subtracting need a common denominator.** You can add halves to halves, not halves to thirds.

- 4 **Cancel factors, never terms.** A factor multiplies the whole numerator or the whole denominator.

- 5 **Keep the minus sign in front.**

$$\frac{-a}{b} = \frac{a}{-b} = -\frac{a}{b}$$

- 6 **Clear a root out of the bottom by multiplying above and below by it.** For a single root, multiply by that root. For $a + \sqrt{b}$, multiply by $a - \sqrt{b}$, its conjugate, because the difference of two squares removes the root.

Rationalise $\frac{5}{\sqrt{2}}$ and $\frac{7}{3 + \sqrt{5}}$

$$\frac{5}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{5\sqrt{2}}{2}$$

One root on the bottom: multiply above and below by it.

$$\frac{7}{3 + \sqrt{5}} \times \frac{3 - \sqrt{5}}{3 - \sqrt{5}}$$

A sum: multiply by the conjugate.

$$= \frac{7(3 - \sqrt{5})}{4}$$

The bottom becomes $9 - 5 = 4$, with no root left.

Simplify $\frac{x^2 - 25}{x + 5}$

$$\frac{(x - 5)(x + 5)}{x + 5}$$

Factor first. Only now is $x + 5$ a factor rather than a term.

$$x - 5$$

Valid for $x \neq -5$, since the original was undefined there.

WATCH OUT

- $\frac{x + 5}{x}$ is not 5. The x on top is a term, so it does not cancel.
- Test any cancellation with a number. At $x = 1$, $\frac{1 + 5}{1} = 6$, not 5.
- Adding numerators and denominators separately is the single most expensive habit on this list.

LEARN IT ON KHAN ACADEMY

Video Dividing fractions by fractions

Arithmetic, unit 15: Dividing fractions

PRACTISE IT ON KHAN ACADEMY

Exercise Adding and subtracting · Add and subtract fractions

Arithmetic, unit 13: Adding and subtracting fractions

Exercise Dividing fractions

Arithmetic, unit 15: Dividing fractions

0.5 Linear equations

Equivalent equations. Two equations are equivalent when they have exactly the same solutions. Solving is nothing more than walking from the equation you were given to an equivalent one simple enough to read.

What preserves equivalence. Adding or subtracting the same thing on both sides. Multiplying or dividing both sides by the same **non-zero constant**. Those are safe, always.

What may not. Squaring both sides, and multiplying by an expression that contains the unknown. Both can create solutions the original never had. $x = 3$ squared gives $x^2 = 9$, which also admits $x = -3$. This is why the check at the end is part of the method, not politeness.

- 1 **Whatever you do, do it to both sides.** That is the only rule; the rest is order.
- 2 **Clear brackets first.** $-(x - 4) = -x + 4$.
- 3 **Then clear fractions by multiplying every term by the common denominator.** Every term, including the ones on the right.
- 4 **Collect the unknown on one side and the numbers on the other.**
- 5 **Divide by the coefficient of the unknown.**
- 6 **If the unknown sits in a denominator, multiply through by it, then check.** A value that makes any original denominator zero is not a solution.

Solve $\frac{2x + 1}{3} - \frac{x - 4}{2} = 1$

$$2(2x + 1) - 3(x - 4) = 6$$

Multiply every term by 6, the common denominator.

$$4x + 2 - 3x + 12 = 6$$

Expand. The minus reaches both terms of the second bracket.

$$x + 14 = 6$$

Collect.

$$x = -8$$

Check: $\frac{-15}{3} - \frac{-12}{2} = -5 + 6 = 1$.

WATCH OUT

- Multiplying by the common denominator and forgetting the right-hand side.
- $-3(x - 4)$ is $-3x + 12$. Both signs change.
- With the unknown in a denominator, an answer that makes it zero must be thrown away.

LEARN IT ON KHAN ACADEMY

Video Unknown on both sides · Linear equations with variables on both sides

Algebra 1, unit 2: Solving equations and inequalities

Video With brackets · Linear equations with parentheses

Algebra 1, unit 2: Solving equations and inequalities

Article The method, written out · Multi-step equations review

Algebra 1, unit 2: Solving equations and inequalities

PRACTISE IT ON KHAN ACADEMY

Exercise Unknown on both sides · Equations with variables on both sides

Algebra 1, unit 2: Solving equations and inequalities

Exercise Unknown in a denominator · Rational equations intro

Algebra 2, unit 10: Equations, including rational ones

0.6 Quadratic equations

- 1 Get everything to one side.

$$ax^2 + bx + c = 0, \quad a \neq 0$$

- 2 Read off a , b and c with their signs. The signs are where the marks go.

- 3 Try factoring. If it factors, it is faster. A product is zero only when one of its factors is zero.

- 4 Otherwise use the formula.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- 5 The discriminant answers how many roots before you find them.

$$\Delta = b^2 - 4ac \quad \Delta > 0: \text{two} \quad \Delta = 0: \text{one} \quad \Delta < 0: \text{none in } \mathbb{R}$$

Solve $2x^2 - 7x + 3 = 0$

$$a = 2, \quad b = -7, \quad c = 3$$

Read them off with their signs.

$$\Delta = 49 - 24 = 25$$

Positive, so there are two roots, and 25 is a square so they are tidy.

$$x = \frac{7 \pm 5}{4}$$

Put it in the formula.

$$x = 3 \quad \text{or} \quad x = \frac{1}{2}$$

WATCH OUT

- $b = -7$ gives $b^2 = 49$, positive. Squaring kills the sign.
- $(x - 2)(x + 5) = 4$ tells you nothing. Only a product equal to *zero* splits.
- A negative Δ means no real solution, which in a business question usually means the target is never reached.

LEARN IT ON KHAN ACADEMY

Video Solving quadratics by factoring

Algebra 1, unit 14: Quadratic functions and equations

Video The formula · The quadratic formula

Algebra 1, unit 14: Quadratic functions and equations

Article The method, written out · Solving quadratics by factoring review

Algebra 1, unit 14: Quadratic functions and equations

PRACTISE IT ON KHAN ACADEMY

Exercise Quadratics by factoring

Algebra 1, unit 14: Quadratic functions and equations

Exercise Using the formula · Quadratic formula

Algebra 1, unit 14: Quadratic functions and equations

0.7 Turning a situation into an equation

- 1 **Name the unknown, with its unit.** Write it down. "Let s = the number of soups sold."
- 2 **Write every other quantity in terms of it.** If 80 items sold and s are soups, the rest is $80 - s$.
- 3 **Find the sentence that says two things are equal. That is the equation.**
- 4 **Solve it.** Now it is only algebra.
- 5 **Answer in words, with the unit, and check it against the situation.** A negative count or a price of 0.003 means the setup was wrong, not the arithmetic.

A stall sells rolls at 4.50 and soups at 7.00. It sold 80 items and took 440.00. How

many soups?

Let s = soups, so rolls = $80 - s$

One unknown is enough; everything else follows from it.

$$7s + 4.5(80 - s) = 440$$

The total takings is the sentence that says two things are equal.

$$2.5s + 360 = 440$$

Expand and collect.

$$s = 32$$

32 soups and 48 rolls. Check: $32(7) + 48(4.5) = 224 + 216 = 440$.

Cost, revenue, profit.

$$C(q) = F + vq \quad R(q) = pq \quad P(q) = R(q) - C(q)$$

F is the fixed cost, what you pay before selling anything. v is the variable cost, what each additional unit adds. p is the price. Break-even is where $R(q) = C(q)$, which is the same as $P(q) = 0$.

WATCH OUT

- Starting to write algebra before defining the letter. Half the lost marks are here.
- Mixing units. If one price is in francs, all of them are.
- Answering with a bare number. The question asked how many soups.

LEARN IT ON KHAN ACADEMY**Video A situation into an equation · Linear equation word problems**

Algebra 1, unit 5: Forms of linear equations

PRACTISE IT ON KHAN ACADEMY**Exercise Writing the equation · Writing linear equations word problems**

Algebra 1, unit 5: Forms of linear equations

Exercise Writing and solving · Linear equations word problems

Algebra 1, unit 5: Forms of linear equations

Exercise Two-step problems · Two-step equations word problems

Algebra basics, unit 3: Linear equations and inequalities

0.8 Inequalities and percentages**1 An inequality solves like an equation, with one exception.**

2 Multiplying or dividing by a negative flips the sign.

$$-2x > 6 \quad \Rightarrow \quad x < -3$$

3 A double inequality: do the same thing to all three parts.

$$2 \leq 3x - 1 \leq 8 \quad \Rightarrow \quad 1 \leq x \leq 3$$

4 A percentage is a multiplier. That is the whole idea.

$$p\% \text{ of } A = \frac{p}{100} \cdot A$$

5 Up by $p\%$ multiplies; down by $p\%$ multiplies too.

$$\times \left(1 + \frac{p}{100}\right) \quad \times \left(1 - \frac{p}{100}\right)$$

6 Successive changes multiply. They never add.

A price rises 20%, then falls 20%. What is the net change?

$$\times 1.20 \quad \text{then} \quad \times 0.80$$

Each change is a multiplier on what came before.

$$1.20 \times 0.80 = 0.96$$

Multiply the multipliers.

a 4% fall

Not zero. The fall is taken on the larger price.

WATCH OUT

- $-2x > 6$ gives $x < -3$. Dividing by the negative flips it.
- Up 20% then down 20% is not back where you started.
- A percentage point is not a percent. From 4% to 6% is two points and a 50% rise.

LEARN IT ON KHAN ACADEMY**Video Multi-step inequalities**

Algebra 1, unit 2: Solving equations and inequalities

Video Percentages · Percent problems

Pre-algebra, unit 4: Percentages

Article Double inequalities · Compound inequalities review

Algebra 1, unit 2: Solving equations and inequalities

PRACTISE IT ON KHAN ACADEMY

Exercise Multi-step linear inequalities

Algebra 1, unit 2: Solving equations and inequalities

Exercise Percentages · Percent word problems

Pre-algebra, unit 4: Percentages

0.9 Systems of linear equations

- 1 **Two unknowns need two independent equations.** Independent means neither is a multiple of the other.
- 2 **Substitution: make one equation say $x = \text{something}$, then put that into the other.** The other one.
Putting it back into the equation you rearranged gives $0 = 0$.
- 3 **Elimination: scale one or both so a letter cancels when you add or subtract.**
- 4 **If the letters vanish and a false statement is left, there is no solution.** $0 = 5$. The lines are parallel.
- 5 **If the letters vanish and a true statement is left, there are infinitely many.** $0 = 0$. It was the same line twice.
- 6 **Check the pair in both equations.** One equation cannot catch an error.

Solve $3x + 2y = 19$ and $x - y = 3$

$$x = y + 3$$

Rearrange the simpler of the two.

$$3(y + 3) + 2y = 19$$

Substitute into the *other* equation.

$$5y = 10, \quad y = 2$$

Expand and collect.

$$x = 5, \quad y = 2$$

Check both: $15 + 4 = 19$ and $5 - 2 = 3$.

Where this is used. Demand falls as the price rises; supply rises as the price rises. Each is a line in price and quantity, and the market settles where the two meet. Finding that point is solving a system of two equations, which is why the two ideas are taught together.

Demand $p = 100 - 2q$, supply $p = 20 + 3q$. Find the equilibrium.

$$100 - 2q = 20 + 3q$$

At equilibrium the two prices are the same.

$$80 = 5q$$

Collect.

$$q = 16$$

The quantity the market clears at.

$$p = 100 - 32 = 68$$

And the price. Check against supply: $20 + 3(16) = 68$.

WATCH OUT

- Substituting back into the equation you just rearranged. It always gives $0 = 0$.
- Subtracting equations and forgetting that the signs on the right change too.
- Checking in one equation only. A slip survives that.

LEARN IT ON KHAN ACADEMY

Video By substitution · Solving systems of equations with substitution

Algebra 1, unit 6: Systems of equations

PRACTISE IT ON KHAN ACADEMY

Exercise By substitution · Systems of equations with substitution

Algebra 1, unit 6: Systems of equations

Exercise By elimination · Systems of equations with elimination

Algebra 1, unit 6: Systems of equations

0.10 Test yourself

Do this week's homework on the course platform, then check it against the answers when they are posted. That is the test. Everything above is how to be ready for it.

Student notes, not official course material.